

Unit 1.9 The logarithmic function. p62-66

Note Title

2015/02/11

Two questions:

Is the function $y = e^x$ one-to-one?

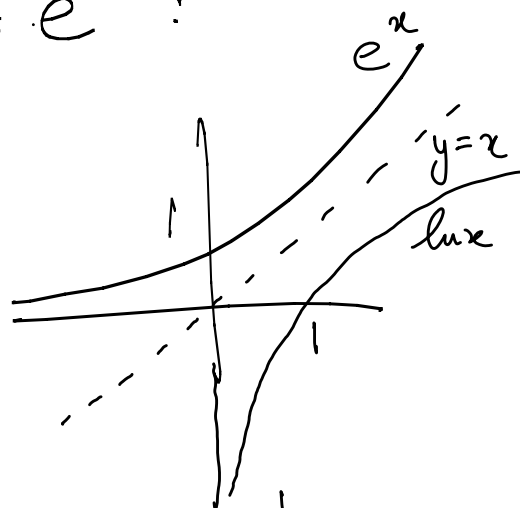
Does $y = e^x$ have an inverse?

Finding the inverse of $y = e^x$:

$$y = e^x$$

For the inverse: $x = e^y$

$$\Rightarrow y = \log_e x = \ln x$$



Eg 1. $y = e$

$$\Rightarrow 3 = \ln y = \ln(e^3)$$

$$e^1 = 2.718$$

$$e^2 = 7.38$$

$$e^3 = 20.08$$

$$e^{13} \approx 1\,000\,000$$

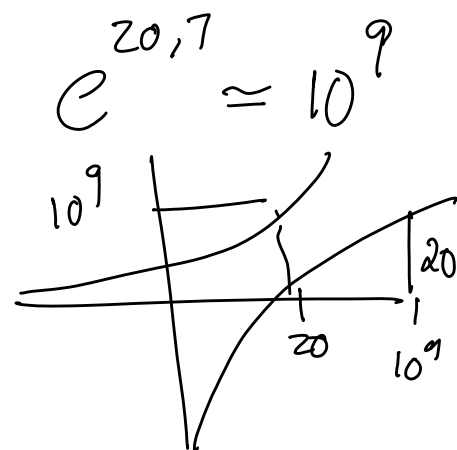
$$2. \ln(7.38) = 2$$

$$3. \ln(8) \approx 2.1$$

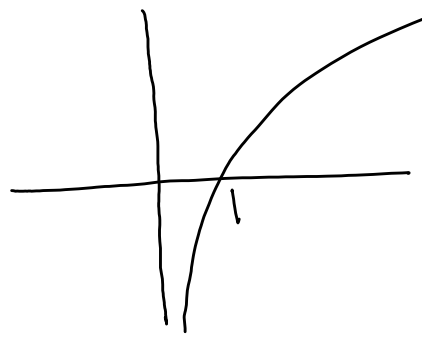
$$4. \ln(25) \approx 3.2$$

$$5. \ln(1\,000\,000) \approx 13$$

$$6. \ln 1\,000\,000\,000 = 20.7$$



Graph of $\ln x$:

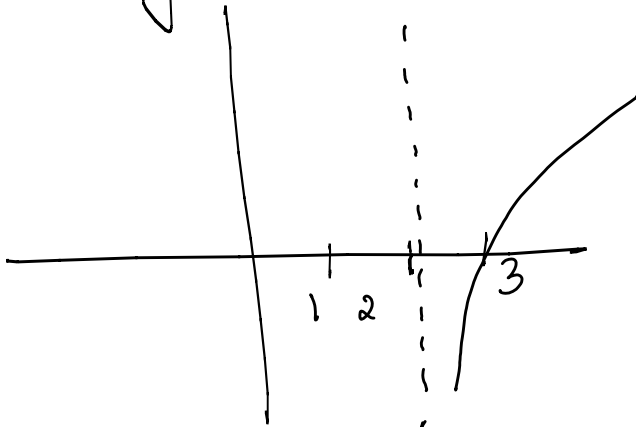


Domain: $(0, \infty)$

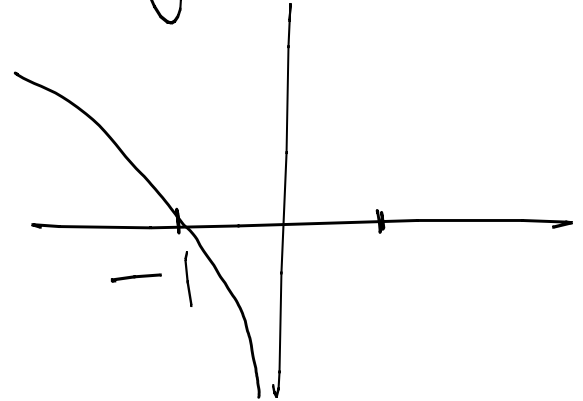
Range: $\mathbb{R}$.

Graphs:

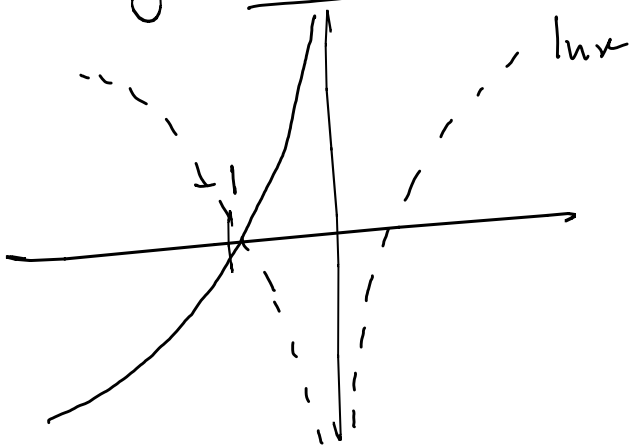
1. $y = \ln(x-2)$



2. $y = \ln(-x)$



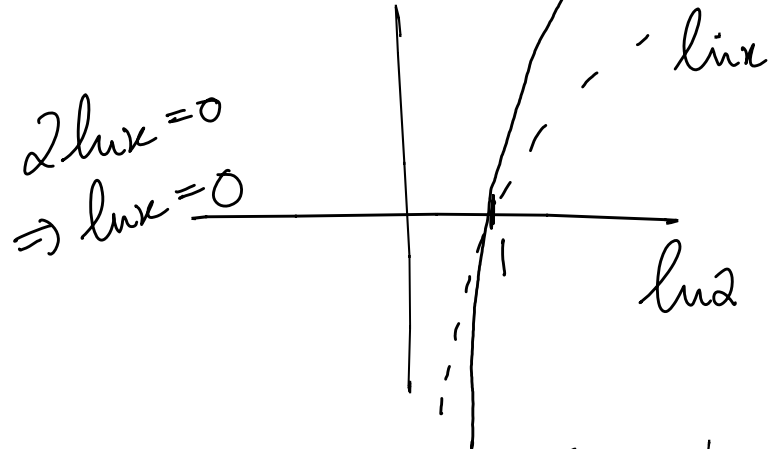
3. $y = -\ln(-x)$



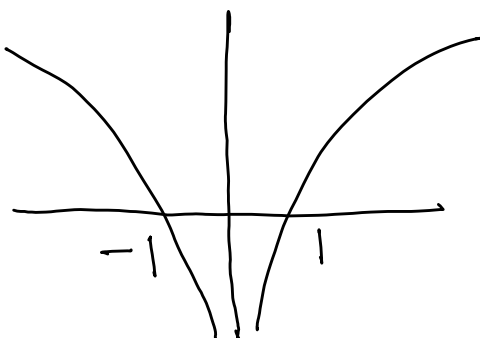
4. $y = 2 \ln x$

$$2 \ln x = 0$$

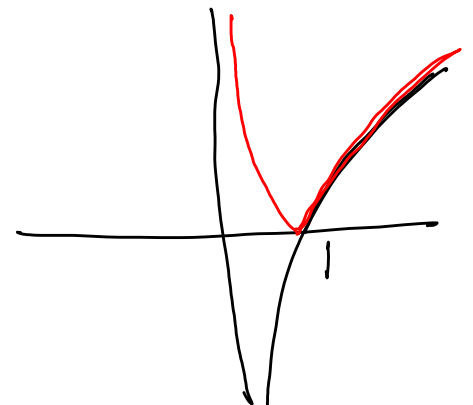
$$\Rightarrow \ln x = 0$$



5. $y = \ln|x|$
 $= \begin{cases} \ln x & \text{if } x > 0 \\ -\ln(-x) & \text{if } x < 0 \end{cases}$



6. $y = |\ln x|$
 $= \begin{cases} \ln x, & \ln x \geq 0 \\ -\ln x, & \ln x < 0 \end{cases}$



Logarithm laws

$$\ln(xy) = \ln x + \ln y.$$

$$\ln\left(\frac{x}{y}\right) = \ln x - \ln y$$

$$\ln(x^n) = n \ln x$$

$$\ln(e^x) = x \quad \text{for } x \in \mathbb{R}$$

$$e^{\ln x} = x \quad \text{for } x > 0.$$

$$\underbrace{\ln 5 - \ln 2}_{\ln 5/2}$$

NB.
?

1. Simplify: $\ln\left(\frac{x^3 \cdot e^{x^2}}{\sqrt{x}}\right)$

$$\begin{aligned} &= \ln x^3 + \ln e^{x^2} - \ln \sqrt{x} \\ &= 3 \ln x + x^2 - \frac{1}{2} \ln x \\ &= 2.5 \ln x + x^2 \end{aligned}$$

2. Solve $2 \ln x - \ln(x-1) = \ln 10 - \ln 2$

$$\Rightarrow \ln x^2 - \ln(x-1) = \ln \frac{10}{2}$$

$$\Rightarrow \ln \frac{x^2}{x-1} = \ln 5$$

$$\Rightarrow \frac{x^2}{x-1} = 5 \Rightarrow x^2 = 5(x-1)$$

$$\Rightarrow x^2 - 5x + 5 = 0$$

$$\Rightarrow x = \frac{5 \pm \sqrt{25 - 20}}{2}$$

Using the logarithmic properties to solve equations:

1.

$$e^{x^2-3} = 2.$$

$$\Rightarrow \ln e^{x^2-3} = \ln 2$$

$$\Rightarrow x^2 - 3 = \ln 2$$

$$\Rightarrow x^2 = 3 + \ln 2$$

$$\Rightarrow x = \pm \sqrt{3 + \ln 2}$$

$$e^{\frac{1}{e}}$$

$$\frac{1}{e} = 0,37$$

$$\ln 2 = 0,69$$

2. $\ln x + \ln(x-1) = 0 \Rightarrow \ln x = -\ln(x-1)$

$$\Rightarrow \ln(x(x-1)) = 0$$

$$\Rightarrow e^{\ln x(x-1)} = e^0 = 1$$

$$\left| e^{\ln \frac{x}{x-1}} = \frac{x}{x-1} \right.$$

$$\Rightarrow x(x-1) = 1,$$

$$\Rightarrow x^2 - x - 1 = 0 \quad \text{etc.}$$

3. $\ln 2x = 4 + \underbrace{3 \ln x}_{\ln x^3}$

$$\Rightarrow \ln 2x - \ln x^3 = 4$$

$$\Rightarrow e^{\ln \frac{2x}{x^3}} = e^4 \Rightarrow \frac{2x}{x^3} = e^4 \quad \text{etc.}$$

$$4. \quad e^x = 9e^{3x}$$

$$\Rightarrow 1 = 9 \frac{e^{3x}}{e^x}$$

$$\Rightarrow \frac{1}{9} = e^{2x}$$

$$\Rightarrow \ln\left(\frac{1}{9}\right) = \ln e^{2x} = 2x$$

$$\Rightarrow x = \frac{1}{2} \ln\left(\frac{1}{9}\right) = \frac{1}{2} (\ln 1 - \ln 9) = \frac{-\ln 9}{2}$$

$$\left| \begin{aligned} \frac{x^2}{2} &= \frac{1}{e^4} \\ \Rightarrow x^2 &= \frac{2}{e^4} \\ \Rightarrow x &= \pm \sqrt{\frac{2}{e^4}} \end{aligned} \right.$$

5. Find the inverse of $f(x) = 1 + e^{-x}$
Sketch f and f^{-1} .

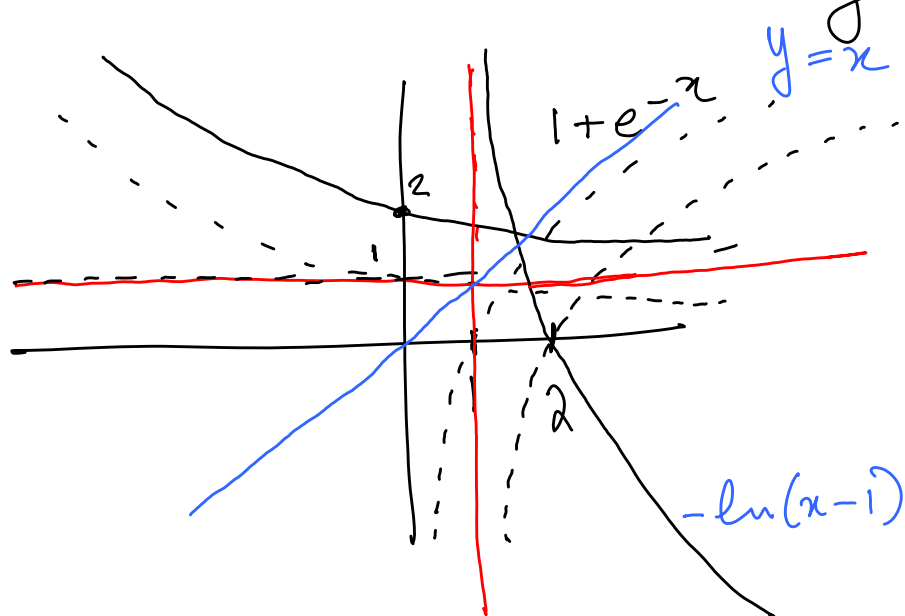
$$y = 1 + e^{-x}$$

$$x = 1 + e^{-y}$$

Inverse:

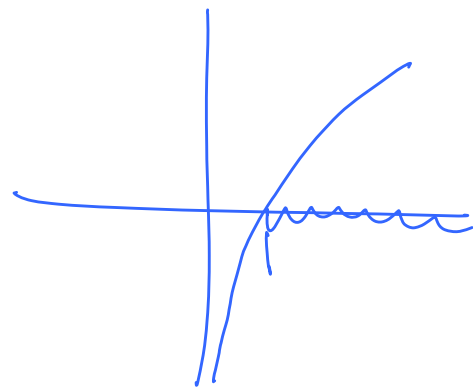
$$\Rightarrow e^{-y} = x - 1 \Rightarrow -y = \ln(x-1)$$

$$\Rightarrow y = -\ln(x-1)$$

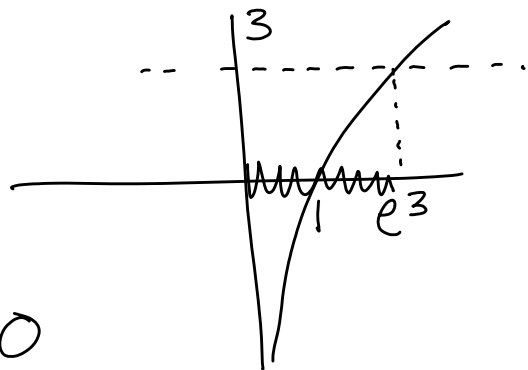


Inequalities

1. Solve: $\ln x > 0$
 $\Rightarrow x > 1.$



2. Solve $\ln x < 3$
 $\Rightarrow e^{\ln x} < e^3$
 $\Rightarrow x < e^3, x > 0$
 $x \in (0, e^3).$



3. Solve $1 - \ln(x+1) > 2$

$\Rightarrow \ln(x+1) < -1.$

$\Rightarrow e^{\ln(x+1)} < e^{-1}$

$\Rightarrow x+1 < e^{-1}$

$\Rightarrow x < -1 + e^{-1}$

$\Rightarrow x \in (-1, -1 + \frac{1}{e})$

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For $\ln(x+1)$:

$x+1 > 0$

$\Rightarrow x > -1.$

Lets talk about domains:

Domain of:

1. $\ln x \Rightarrow \text{Domain } (0, \infty)$

2. $\ln x^2 \Rightarrow \text{Domain } (-\infty, \infty) \setminus \{0\}$

3. $\ln(1-x) \Rightarrow \text{Domain } (-\infty, 1), \quad \begin{matrix} 1-x > 0 \\ x < 1 \end{matrix}$

4. $\ln(4-x^2)$
 $\Rightarrow 4-x^2 > 0 \Rightarrow x^2 < 4$
 $\Rightarrow -2 < x < 2$
 $\Rightarrow x \in (-2, 2)$

5. $\frac{1}{\ln x}$

$$\begin{aligned} \ln x &\neq 0 \quad \text{and} \quad x > 0 \\ \Rightarrow x &\neq 1 \quad \text{and} \quad x > 0 \\ \Rightarrow x &\in (0, \infty) \setminus \{1\} \end{aligned}$$

6. $\frac{1}{1 - \ln(x-1)}$
 $1 - \ln(x-1) \neq 0 \quad \text{and} \quad x-1 > 0$
 $\Rightarrow \ln(x-1) \neq 1 \quad \text{and} \quad x > 1$
 $\Rightarrow x-1 \neq e \quad \text{and} \quad x > 1$
 $\Rightarrow x \in (1, \infty) \setminus \{1+e\}$